

Exercises: Calculating Sets

Mathematics for AI - June 2026

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1 Exercise 1.1: Power set relationship and proof

Confirming the relationship between $|\wp(A)|$ and $|A|$.

In the table below I test a few small sets and write out their subsets. Counting these then gives $|\wp(A)|$.

A. Trying small examples:

A	$ A $	subset	$ \wp(A) $
$\emptyset$	0	$\emptyset$ is its only subset	1
$\{a\}$	1	$\emptyset; \{a\}$	2
$\{a, b\}$	2	$\emptyset; \{a\}, \{b\}; \{a, b\}$	4
$\{a, b, c\}$	3	$\emptyset; \{a\}, \{b\}, \{c\}; \{a, b\}, \{a, c\}, \{b, c\}; \{a, b, c\}$	8
$\{a, b, c, d\}$	4	$\emptyset; \{a\}, \{b\}, \{c\}, \{d\}; \{a, b\}, \{a, c\}, \{a, d\}, \{b, c\}, \{b, d\}, \{c, d\}; \{a, b, c\}, \{a, b, d\}, \{a, c, d\}, \{b, c, d\}; \{a, b, c, d\}$	16

The example shows the following relationship:

$$|\wp(A)| = 2^{|A|}$$

Proposition 1. For every finite set A , $|\wp(A)| = 2^{|A|}$

Proof. I will use induction on n which is $n = |A|$.

Base case where ($n = 0$). Here we use the empty set $A = \emptyset$. Its only subset is itself, so $\wp(\emptyset) = \{\emptyset\}$ and $|\wp(\emptyset)| = 1$ which is $= 2^0$.

Inductive step. Assume that the proposition holds for every set of size n ; such that any set with cardinality n has exactly 2^n subsets. Now consider a set A with $|A| = n + 1$. Choose some element $z \in A$ and set $A' = A \setminus \{z\}$, so that $|A'| = n$.

All subsets $S \subseteq A$ fall into exactly one of two cases:

- $z \notin S$. Then $S \subseteq A'$, so S is a subset of A' . In contrast, every subset of A' is also a subset A , which does not contain z . So the subsets of A with no element z are exactly the subsets of A' and from the inductive hypothesis there are 2^n of them.
- $z \in S$. Define $S' = S \setminus \{z\}$, so that $S' \subseteq A'$ and $S = S' \cup \{z\}$. If we take z away and then add it back, these are inverse operations that set the corresponding pairs (a bijection) between the subsets of A with z and the subsets of A' . Since these match perfectly, each collection will have the same number of subsets, which from the inductive hypothesis we know to be 2^n .

These two groups are disjoint (one excludes z , the other contains it) and together they give every subset of A . Hence

$$|\wp(A)| = 2^n + 2^n = 2 \cdot 2^n = 2^{n+1} = 2^{|A|}.$$

Based on mathematical induction, $|\wp(A)| = 2^{|A|}$ for every finite set A . □

Another way of considering this proof There is a quicker way to get to the same count by counting choices. To form a subset we make, independently for each of the $|A|$ elements, a binary choice (*in* or *out*). That is $2^{|A|}$ possible subsets.

2 Exercise 1.2: distributivity law of intersection over union

Proposition 2. *For all sets A , B and C ,*

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Proof. Sets that have the same elements are equal. If we have an arbitrary element z , then if we can prove it is a member of the left hand set if and only if it is a member of the right hand set, then we have proven that both sides are equal.

$$\begin{aligned} z \in A \cap (B \cup C) &\iff z \in A \wedge z \in (B \cup C) && \text{(definition of intersection } \cap) \\ &\iff z \in A \wedge (z \in B \vee z \in C) && \text{(definition of union } \cup) \\ &\iff (z \in A \wedge z \in B) \vee (z \in A \wedge z \in C) && \text{(the distributive law for } \wedge \text{ over } \vee) \\ &\iff (z \in A \cap B) \vee (z \in A \cap C) && \text{(definition of intersection } \cap) \\ &\iff z \in (A \cap B) \cup (A \cap C) && \text{(definition of union } \cup). \end{aligned}$$

Since the z we used was arbitrary, the above holds for all elements, the two sets thus have the same elements and are equal. □

Note: a key step above is the distributive law step. Here we mirror the property from arithmetic where $a(b + c) = ab + ac$. In the above case, *and* distributes over *or* with $\cap$ acting like multiplication and $\cup$ acting like addition.

3 Exercise 1.3: Prove that $|A \cup B| = |A| + |B| - |A \cap B|$

Proposition 3. *For all sets A and B ,*

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Note: the proof below relies on the following: if two sets A and B do not overlap so $A \cap B = \emptyset$ then $|A \cup B| = |A| + |B|$

Proof. $(A \cup B)$ can be expanded into three parts:

- $A \setminus B$
- $(A \cap B)$
- $B \setminus A$

The above three parts are pairwise disjoint, meaning that no two of them share any elements. Their union is $A \cup B$ since every element of $A \cup B$ is in one of the three parts.

Next we apply the additive principle but we have to do this twice since we have 3 parts but the principle only applies to two things:

$$|A \cup B| = |A \setminus B| + |A \cap B| + |B \setminus A|. \quad (1)$$

Now we can consider what $|A \setminus B|$ is: it is the same as considering all of A minus the intersection of A and B, and the same principle applies for B.

$$|A \setminus B| = |A| - |A \cap B|, \quad (2)$$

$$|B \setminus A| = |B| - |A \cap B|. \quad (3)$$

Equation (2) and (3) can then be substituted into equation (1):

$$|A \cup B| = |A| - |A \cap B| + |A \cap B| + |B| - |A \cap B|$$

This simplifies to:

$$|A \cup B| = |A| + |B| - |A \cap B| \quad (4)$$

Equation (4) is exactly the statement of the proposition which completes our proof. $\square$